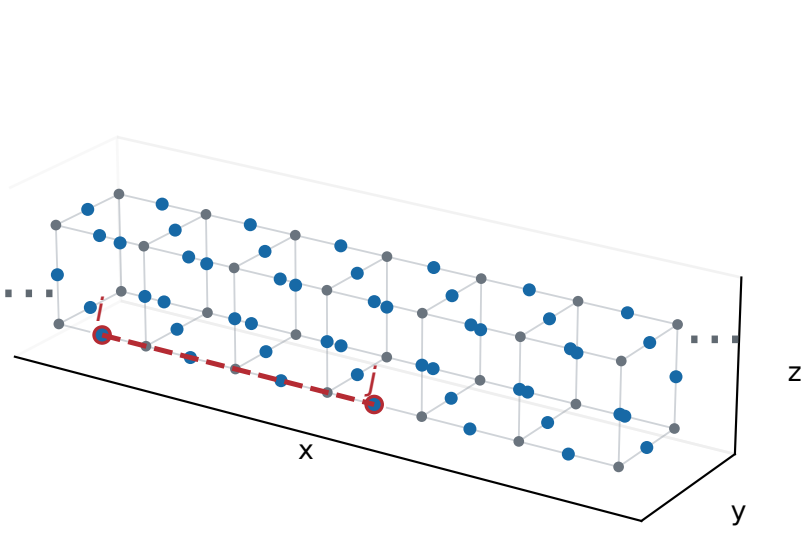
Your lattice is three-dimensional. The previous calculation used a narrow tube.

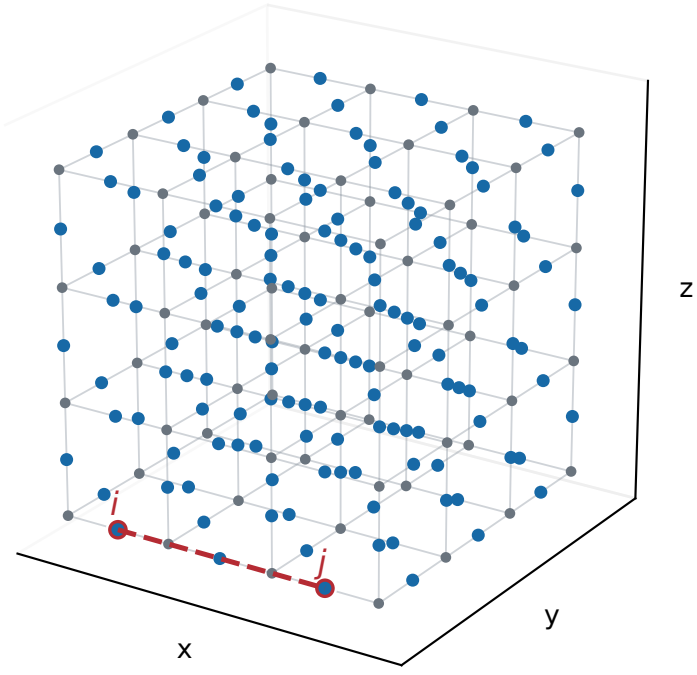
Previous approximation: finite transverse size



$L_x \to \infty$, but $L_y = 2$, $L_z = 2$

Periodic boundaries in y, z; a finite window is shown.

Requested target: the bulk 3D lattice



$L_x, L_y, L_z \to \infty$ together

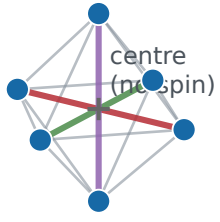
Increase boxes in all three directions; test convergence.

Physical spin 1 at a cubic-bond midpoint

Octahedron centre: no physical spin

Cubic grid: geometric guide, not a singlet bond

One octahedron: six physical spins



Local geometry

Coloured diameters show one allowed three-dimer support.
A trimer instead occupies one triangular face. Adjacent
octahedra share a midpoint spin, which is covered once.

The red pair measures a physical spin correlation along x:
$C(r) = \langle \mathbf{S}_i \cdot \mathbf{S}_{i+r\hat{x}} \rangle$. The surrounding y, z directions still affect it.

Geometry schematic only. Periodic wrapping bonds are omitted. No converged bulk correlation length is claimed by this figure.